JUUM

STRUMPLATE SYNTH · QUICK START & MANUAL

The PDF manual is generated from this online guide whenever the manual changes.

JUUM is a browser-based synthesizer. Play notes on the large strumplate, shape two oscillator voices, select chords, and explore MIDI clock features.

Make your first sound

1. Open JUUM in a modern browser such as Chrome, Edge, or Safari.
2. Turn up your computer or phone volume.
3. Tap or click the large blue **Strumplate**. Drag across it to play as you move.
4. If you're unsure whether audio is working, press **TEST TONE**. It should play a short A4 note.
5. Press **MAKE NOISE** for a brief noise burst to check the audio output.

Browsers generally require a tap/click before audio can start. If silent, interact with the page once, check device volume and output selection, then refresh.

Voice controls

Main Voice

The primary oscillator defines the main tone. Choose its waveform and adjust Volume, Cutoff, Resonance, Attack, Decay, Sustain, and Release. Use the Main Voice switch to mute or enable this layer.

Sub Voice

The sub oscillator adds a lower layer one octave beneath the played note. Its own waveform, volume, filter, and envelope shape the bass tone. Toggle it off to hear only the Main Voice.

Detune

The Detune knob shifts the Sub Voice by cents relative to its normal pitch. Start near 0; small positive or negative settings create a thicker, beating sound. Larger shifts sound more obviously out of tune.

Knobs

Drag a knob vertically to change its value. On touchscreens, drag with a finger. Use the displayed number as a guide; turn a control back toward its starting value if you lose track.

Chords & arpeggiator

Choose a root note and scale in Global. The chord buttons select the harmony. The arpeggiator turns the selected chord into a repeating pattern.

- **Rate:** notes per second.
- **Octaves:** how many octaves the pattern spans.
- **Swing:** adds rhythmic lilt.
- **Gate:** controls how long each arpeggiated note sustains.

- **Mode:** Up, Down, Up/Down, or Random.

Use the ARP control to start or stop the pattern. The strumplate is the hands-on playing surface.

Effects

Reverb adds a sense of space. **Delay** repeats the sound; Delay Time sets the spacing between repeats. **FX Volume** controls the overall effects return level, while **Master** controls the overall synth output. The Reverb and Delay switches bypass their effects while leaving their knob settings available.

Effect amounts at zero may sound like an effect is off even when its switch is enabled.

Preset Memory, Waveform & Sequencer

Preset Memory

Use Save Preset to name and store the current sound, select it from the list, then choose Load or Delete. Presets are stored in this browser’s local storage and do not automatically sync between devices.

Live Waveform

The waveform display visualizes the audio output signal. Interact with the synth or press TEST TONE to activate the audio engine.

16-Step Sequencer

1. Choose a chord in Chord Select and set your preferred Root note and Scale in Global.
2. Click numbered steps 01–16 to toggle them on or off. Enabled steps light up.
3. Try steps **01, 05, 09, and 13** for four evenly spaced hits.
4. Set the sequencer BPM, press PLAY, and press STOP when finished.

Important: Notes are currently generated automatically from the selected chord. Individual pitches cannot yet be assigned to each step. If you hear no sound, enable at least one step, select a chord, make sure a voice is enabled, and check Master/FX/voice volume.

MIDI & clock

JUUM's Web MIDI support depends on browser and device support. On many desktop browsers, MIDI requires a secure HTTPS page and a compatible MIDI device.

- **MIDI:** requests MIDI access and checks for available outputs.
- **Play / Stop:** starts and stops outgoing MIDI clock; it does not start the internal audio synthesizer.
- **Send MIDI clock:** enables outgoing clock messages.
- **MIDI Thru:** passes supported incoming notes to the selected output.
- **Receive clock:** enables incoming clock detection where supported.
- **Send notes:** enables note messages to MIDI output.

Troubleshooting

No sound?

- Click TEST TONE once to unlock browser audio.
- Raise computer/device volume and confirm the correct audio output is selected.
- Check the Master and Main Voice Volume knobs.
- Make sure at least one voice is enabled.
- Try Chrome, Edge, or Safari and reload the page.

Controls hard to use on a phone?

Rotate to portrait or landscape and zoom only if needed. Drag knobs vertically, and tap the strumplate. Keep the browser tab active while playing.

What is still experimental?

JUUM is an evolving project. Browser audio and MIDI behavior varies by device. The strumplate is a touch/mouse control, not a physical harp sensor. Some features may behave differently as the project develops.